(5) Reinterpret_cast

int a = 65;
char *p = reinterpret_cast<char*>(&a);
# low level conversation.

Problem statement 1: Reference variable basic.
Problem statement 2: swap using Reference
Problem statement 3: type casting in Polymorphism

Create: • a base class shape with virtual func
• Derived class Circle.

Task: • Create base pointer pointing to circle
• using dynamic_cast to convert it into a derived pointer
• call derived pointer

Day 45 (Abstraction class)

Pure virtual function is a virtual function with no body in the base class

Syntax: virtual void show() = 0;
=0 make it a pure virtual

Why use? • To force derived classes to defined the func
• Create a common inheritance

Abstract Base Class: A class that contain atleast virtual function is called an abstract base c

Problem statement 1: Shape area system.

Create a abstract base class with pure virtual f

Derive rectangle and circle and implement area

Problem statement 2: Animal sound system

Problem statement 3: Employee salary system.

Full time, intern as a derived class.

Salary as function.

Important rule

We cannot creat a object of abstract class.

Shape s ;

But Pointer is allowed

Shape * ptr ;

Example

```
# include <iostream>
using namespace std;
Class Shape {
   Public :
      virtual void area () = 0; // Pure virtual.
};

Class Rectangle : public shape {
  Public :
    void area () {
          Cout << "Rectongee area" << endl;
    }
};

Class Circle : Public Shape {
   Public :
     void area () {
          Cout << " Circle Area " << endl;
   }
};

int main () {
   Shape * ptr ;
   Ptr = new Rectangle ();
     Ptr -> area () ;
   Ptr = new Circle ();
     Ptr -> area ();                return 0 ;
                                    }
```